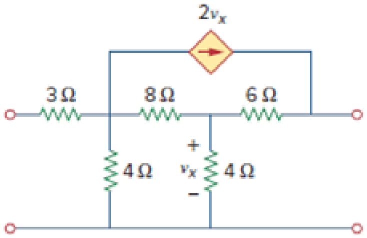
Chapter 19, Problem 15PP

Problem

Obtain the h parameters for the network in Fig. 19.51 using *PSpice.*

Figure 19.51



Step-by-step solution

Step 1 of 6

The hybrid parameters

$V_1 = h_1 I_1 + h_{12} V_2$

$I_2 = h_{21} I_1 + h_{22} V_2$

From the general equation, we get

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0}$$

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0}$$

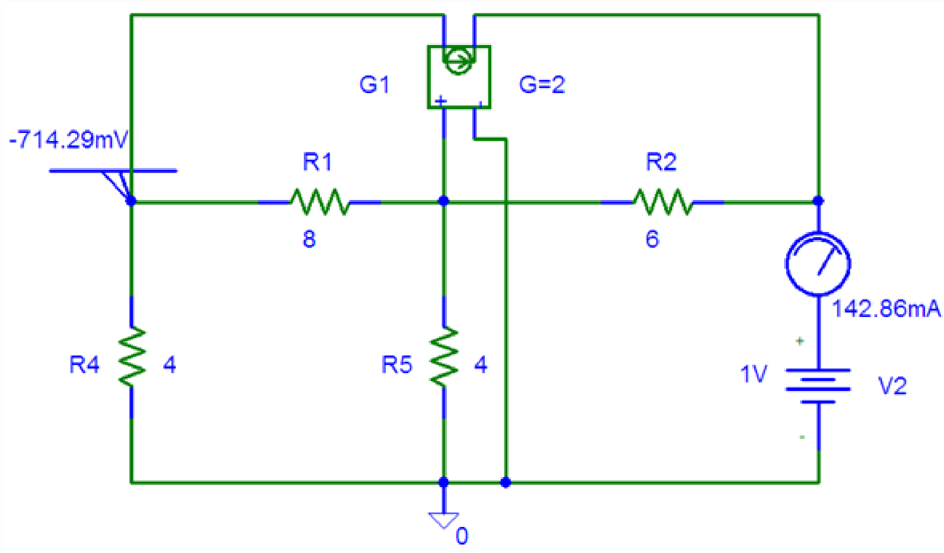
h_{11} And h_{21} can be found by setting $V_2 = 0$. Also by setting $I_1 = 1\text{A}$,

h_{11} becomes $\frac{V_1}{1}$ While

h_{21} becomes $\frac{I_2}{1}$.

We insert a 1A dc current source IDC to take care of $I_1 = 1\text{A}$, the pseudo component VIEWPOINT to display V_1 and pseudo component IPROBE to display I_2. Thus, the schematic is

Step 2 of 6



Step 3 of 6

Save the schematic, then run pspice by selecting analysis/Simulate and note the values displaced on the pspice components.

$$h_{11} = \left.\frac{V_1}{1}\right|_{V_2=0}$$

$$h_{11} = 4.238\Omega$$

Actual current direction is from down to top but here we got current direction from top to down. So we should take the same magnitude with

$$h_{21} = \left.\frac{I_2}{1}\right|_{V_2=0}$$

$$h_{21} = -0.6190$$

Step 4 of 6

Similarly, from the general equation

$$h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0}$$

$$h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0}$$

We obtain h_{12} and h_{22} by open-circuiting the input port $(I_1 = 0)$. By making $V_2 = 1\text{V}$

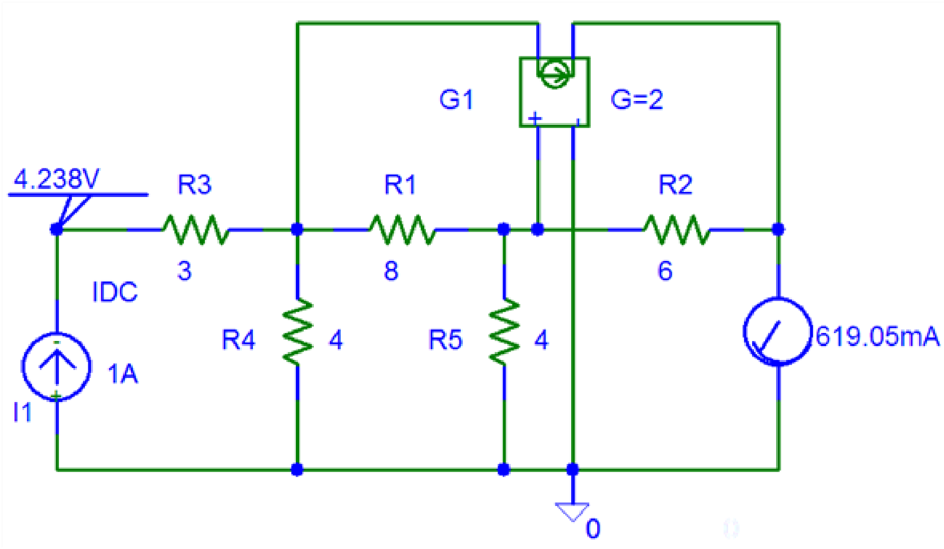
h_{12} becomes $\frac{V_1}{1}$ while

h_{22} becomes $\frac{I_2}{1}$.

Thus, the schematic with a 1V dc voltage source VDC inserted at the output terminal to take care of $V_2 = 1\text{V}$.

The pseudo component VIEWPOINT and IPROBE are inserted to display to values of V_1 and I_2 respectively, the 3Ω resistor is ignored because the input port is open circuited and pspice will not allow such.

Step 5 of 6



Step 6 of 6

Save the schematic, after simulating the schematic, we obtain the values displayed on the. Thus,

$$h_{12} = \frac{V_1}{1}\bigg|_{I_1=0}$$

$$h_{12} = \frac{-0.7143}{1}$$

$$h_{12} = -0.7143$$

$$h_{12} = \frac{I_2}{1}\bigg|_{I_1=0}$$

$$h_{22} = \frac{-0.1429}{1}$$

$$h_{22} = -0.1429\,\text{S}$$

Therefore, the h parameters for the network are

$$h_{11} = 4.238\Omega \quad h_{12} = -0.7143$$

$$h_{21} = -0.6190 \quad h_{22} = -0.1429\,\text{S}$$